

Measurement Auditing for Learned Ghost Imaging: Certificates, Limits, and Prior-Supplied Content

[Authors omitted for review]

Abstract—Learned ghost-imaging reconstructions can be visually plausible while still leaving unclear which image content is supported by the recorded bucket measurements and which content is supplied by the prior. This paper develops a range-null view of this question. For a sensing matrix $A \in \mathbb{R}^{m \times n}$, $m \ll n$, the row-space projector $P_R = A^\dagger A$ and null-space projector $P_0 = I - A^\dagger A$ give the unique orthogonal decomposition $x = P_R x + P_0 x$. The measurement record constrains only $P_R x$; $P_0 x$ is invisible to A . This single geometric fact separates measurement accountability from image quality, and it also exposes a limit: consistency with the bucket measurements is not correctness of the unmeasured image content. We then attach to any reconstructor a test-time measurement audit, $\Pi_y^\lambda(v) = v - B_\lambda(Av - y)$ with $B_\lambda = A^\top(AA^\top + \lambda I)^{-1}$, and prove that its residual contracts by the singular-mode factor $\lambda/(\lambda + \sigma_i^2)$. Experiments show that the same audit reduces measurement residuals for BP, Tikhonov, CS-TV, and learned outputs while moving PSNR only slightly for trained outputs. A GAN branch is used only as a representative generative prior: it illustrates controlled prior-supplied detail under the same audit, not a claim of state-of-the-art image quality or superiority over diffusion models.

Index Terms—Ghost imaging, inverse problems, measurement consistency, null space, range space, learned reconstruction, GAN, certification.

I. INTRODUCTION

Low-rate ghost imaging and single-pixel imaging acquire a small number of bucket measurements from structured illumination patterns [CITE]. At sampling rates such as 5–10%, the inverse problem is deeply underdetermined: many images agree with the same measurement vector. Modern learned priors, including adversarial and other generative priors, can fill the missing visual structure and produce reconstructions that look plausible. The scientific question is therefore not only whether the image is visually good. It is also: which part of the image is supported by the recorded measurement, and which part was supplied by the prior?

Standard image metrics do not answer this question. PSNR and SSIM require a ground-truth image, which is unavailable in deployment, and they mix together measured and unmeasured error. A high PSNR image can still have a measurement residual, while a measurement-consistent image can still be semantically wrong in its unmeasured component. A useful reconstruction pipeline for undersampled imaging should therefore report a ground-truth-free measurement-accountability certificate in addition to image-quality metrics.

This work takes a deliberately geometric route. Let $A \in \mathbb{R}^{m \times n}$, $m \ll n$, be the ghost-imaging measurement operator. The Moore–Penrose row-space projector $P_R = A^\dagger A$ and null-space projector $P_0 = I - A^\dagger A$ split every image into measured and unmeasured components. This decomposition

is not a modeling assumption; it is the linear algebra of the measurement operator. The row component $P_R x$ is the part seen by the bucket measurements, while $P_0 x$ is exactly invisible to them.

The paper’s contributions are:

- We use the range-null decomposition to give a strict separability statement for measurement accountability and prior-supplied image content. The measurement $y = Ax$ constrains $A(P_R x)$ and never constrains $A(P_0 x)$.
- We define a plug-in test-time audit Π_y^λ that can be applied after any reconstructor. We prove its singular-mode residual contraction exactly and report the corresponding measurement certificate.
- We state the boundary of this certificate. The audit certifies agreement with the recorded measurements; it does not certify that the unmeasured component is true. We empirically map this boundary through feasible hallucinations, operator-drift and wrong-measurement probes, and posterior-calibration diagnostics.
- We use a GAN branch as a representative generative prior under this audit. The GAN is not the novelty claim and is not presented as a state-of-the-art benchmark result or as a comparison against diffusion models.

All numerical values reported in the tables are from the experiments described in this manuscript.

II. PROBLEM SETUP AND RANGE-NULL DECOMPOSITION

A. Forward Model

We consider the standard linear ghost-imaging model

$$y = Ax + \varepsilon, \quad (1)$$

where $x \in \mathbb{R}^n$ is the vectorized image, $y \in \mathbb{R}^m$ is the bucket-measurement vector, $A \in \mathbb{R}^{m \times n}$ is the measurement matrix, $m \ll n$, and ε is measurement noise. Let the compact singular value decomposition of A be

$$A = U_r \Sigma_r V_r^\top, \quad (2)$$

where $r = \text{rank}(A)$, $U_r \in \mathbb{R}^{m \times r}$ and $V_r \in \mathbb{R}^{n \times r}$ have orthonormal columns, and $\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r)$ with $\sigma_i > 0$.

The Moore–Penrose pseudoinverse is

$$A^\dagger = V_r \Sigma_r^{-1} U_r^\top. \quad (3)$$

Therefore

$$P_R = A^\dagger A = V_r \Sigma_r^{-1} U_r^\top U_r \Sigma_r V_r^\top = V_r V_r^\top, \quad (4)$$

and

$$P_0 = I - P_R = I - V_r V_r^\top. \quad (5)$$

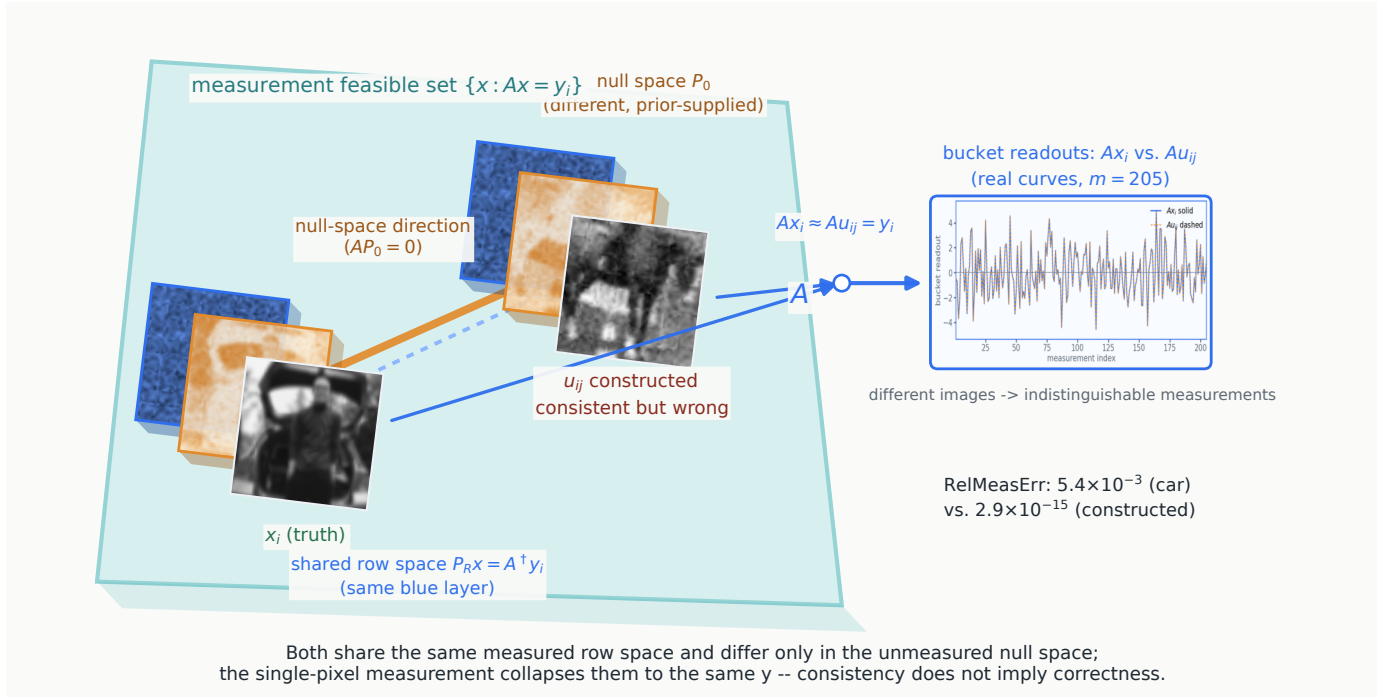


Fig. 1. Range-null decomposition and the ghost-imaging bucket readout for the Rad-5 pair $i = 1789$, $j = 935$. Each image card is drawn as a real three-layer stack: the shared blue row-space layer is the measurement-locked skeleton $P_R x = A^\dagger y_i$, the orange layer is the null-space component, and the top layer is the displayed image. The car truth x_i uses $P_0 x_i$, whereas the constructed image $u_{ij} = x_j - A^\dagger (Ax_j - y_i)$ uses the horse donor's $P_0 x_j$; hence the two cards share the measured row space and differ only in the unmeasured null space. Both are sent through the same single-pixel measurement operator, and the right panel plots the real bucket readout vector $y_i = y[1789]$ ($m = 205$), not a schematic. The car truth has the noisy-record residual $\text{RelMeasErr}(x_i, y_i) = 5.36 \times 10^{-3}$, while the constructed image satisfies $\text{RelMeasErr}(u_{ij}, y_i) = 2.93 \times 10^{-15}$. The displayed u_{ij} is clipped for visualization and is constructed, measurement-consistent, semantically wrong, and not a natural image. Thus measurement-consistent $\neq$ correct $\neq$ natural.

Since $V_r^\top V_r = I_r$, we have

$$P_R^2 = (V_r V_r^\top)(V_r V_r^\top) = V_r (V_r^\top V_r) V_r^\top = V_r V_r^\top = P_R, \quad (6)$$

$$P_R^\top = P_R, \quad (7)$$

so P_R is an orthogonal projector. The same identities give

$$P_0^2 = (I - P_R)^2 = I - 2P_R + P_R^2 = I - P_R = P_0, \quad (8)$$

$$P_0^\top = P_0. \quad (9)$$

Moreover,

$$P_R P_0 = P_R (I - P_R) = P_R - P_R^2 = 0. \quad (10)$$

Thus, for every image x ,

$$x = P_R x + P_0 x, \quad (11)$$

and

$$\langle P_R x, P_0 x \rangle = x^\top P_R^\top P_0 x = x^\top P_R P_0 x = 0. \quad (12)$$

Plain-language interpretation: every image splits uniquely into a measured row-space part and an unmeasured null-space part, and the two parts do not overlap in energy.

B. Measurements Constrain Only the Row Space

We now derive the central separability statement. First,

$$AP_0 = A(I - A^\dagger A) \quad (13)$$

$$= A - AA^\dagger A. \quad (14)$$

The Moore–Penrose identities give $AA^\dagger A = A$, so

$$AP_0 = 0. \quad (15)$$

Similarly,

$$AP_R = A(A^\dagger A) \quad (16)$$

$$= (AA^\dagger A) \quad (17)$$

$$= A. \quad (18)$$

Combining (11), (15), and (18),

$$Ax = A(P_R x + P_0 x) \quad (19)$$

$$= AP_R x + AP_0 x \quad (20)$$

$$= AP_R x. \quad (21)$$

Therefore the measurement record is a function only of $P_R x$.

In the noiseless case, if $Ax = y$, then

$$P_R x = A^\dagger Ax = A^\dagger y. \quad (22)$$

Thus every feasible image with the same measurement y has the same row-space component $A^\dagger y$. Conversely, for any $z_0 \in \text{null}(A)$,

$$A(A^\dagger y + z_0) = AA^\dagger y + Az_0. \quad (23)$$

When $y \in \text{range}(A)$, $AA^\dagger y = y$, and $Az_0 = 0$, so

$$A(A^\dagger y + z_0) = y. \quad (24)$$

The set of noiseless feasible images is therefore

$$\{x : Ax = y\} = A^\dagger y + \text{null}(A). \quad (25)$$

With noise, $A^\dagger y$ is the least-squares row-space estimate induced by the recorded measurement, while the null component remains unobserved by A .

Plain-language interpretation: the bucket measurements determine the measured component and say nothing about the null-space content; the underdetermined part is not a weakness of a particular algorithm, but a geometric fact about A .

III. MEASUREMENT AUDIT AND ITS CONTRACTION CERTIFICATE

A. A Plug-In Audit

Let $v \in \mathbb{R}^n$ be the output of any reconstructor: a backprojection, a variational solver, a neural network, or a generative model. Define

$$B_\lambda = A^\top (AA^\top + \lambda I)^{-1}, \quad \lambda > 0, \quad (26)$$

and the test-time audit

$$\Pi_y^\lambda(v) = v - B_\lambda(Av - y). \quad (27)$$

The audit uses only A , y , and the candidate output v ; it does not retrain the reconstructor.

The update direction $B_\lambda(Av - y)$ lies in $\text{range}(A^\top)$, which is the row space of A . Hence

$$P_0 B_\lambda(Av - y) = 0, \quad (28)$$

and therefore

$$P_0 \Pi_y^\lambda(v) = P_0 v. \quad (29)$$

The audit changes only the measurement-visible component.

Plain-language interpretation: the audit can repair bucket consistency, but it does not rewrite the prior-supplied null-space image content.

B. Residual Contraction

Let

$$r(v) = Av - y \quad (30)$$

be the measurement residual before audit. Applying A to (27) gives

$$A \Pi_y^\lambda(v) - y = A(v - A^\top (AA^\top + \lambda I)^{-1}(Av - y)) - y \quad (31)$$

$$= Av - y - AA^\top (AA^\top + \lambda I)^{-1}(Av - y) \quad (32)$$

$$= [I - AA^\top (AA^\top + \lambda I)^{-1}] r(v). \quad (33)$$

Now use

$$AA^\top (AA^\top + \lambda I)^{-1} = [(AA^\top + \lambda I) - \lambda I] (AA^\top + \lambda I)^{-1} \quad (34)$$

$$= I - \lambda (AA^\top + \lambda I)^{-1}. \quad (35)$$

Substituting (35) into (33),

$$A \Pi_y^\lambda(v) - y = [I - (I - \lambda (AA^\top + \lambda I)^{-1})] r(v) \quad (36)$$

$$= \lambda (AA^\top + \lambda I)^{-1}(Av - y). \quad (37)$$

To see the mode-wise certificate, write

$$AA^\top = U_r \Sigma_r^2 U_r^\top \quad (38)$$

on the nonzero left singular subspace. If $r(v) = \sum_{i=1}^r \alpha_i u_i$ in that subspace, then

$$A \Pi_y^\lambda(v) - y = \lambda U_r (\Sigma_r^2 + \lambda I)^{-1} U_r^\top \sum_{i=1}^r \alpha_i u_i \quad (39)$$

$$= \sum_{i=1}^r \frac{\lambda}{\sigma_i^2 + \lambda} \alpha_i u_i. \quad (40)$$

Thus each measured singular mode contracts by

$$c_i(\lambda) = \frac{\lambda}{\lambda + \sigma_i^2}. \quad (41)$$

Plain-language interpretation: the audit supplies a computable certificate: after one audit, every measured residual mode is reduced by a known factor determined only by A and λ .

The contraction check confirms this formula in float64 arithmetic for $k = 1$: Rad-5 has maximum float64 mode deviation 1.04×10^{-10} with $\sigma_{\min} = 3.476$ and $\sigma_{\max} = 5.454$, while Scr-5 has maximum float64 mode deviation 2.29×10^{-12} with $\sigma_{\min} = \sigma_{\max} = 1.000$. Repeated float32 pipeline audits hit a solver floor, so multi-step claims must be stated with arithmetic precision in mind.

C. Post-Hoc Evidence Across Reconstructors

Table I reports representative post-hoc audit rows. The audit is applied after the reconstructor output is produced. The rows include analytic backprojection (BP), Tikhonov, a small-subset CS-TV sanity check, and learned outputs. CS-TV is explicitly small-subset evidence with $n = 8$ for the rows shown.

These rows show the intended role of the audit: large reductions in RelMeasErr can be obtained without retraining and with very small PSNR movement for the trained outputs. The quantitative claim here is restricted to the rows in Table I; the geometric PSNR headroom argument is developed separately in Section IV.

The parameter λ should be interpreted with the noise floor. Sending λ to zero is a hard projection in exact arithmetic, but a noisy measurement should not be over-interpreted as exact truth. Selecting a Morozov-style operating point is outside the scope of the present experiments.

IV. SEPARABILITY OF IMAGE QUALITY AND MEASUREMENT ACCOUNTABILITY

A. Orthogonal Error Decomposition

Theorem 1 (hard-projection PSNR ceiling): Let $\hat{x}$ be any pre-audit reconstruction and define the pre-audit error

$$e = e_{\text{pre}} = \hat{x} - x. \quad (42)$$

Using the projectors from Section II,

$$e = P_R e + P_0 e. \quad (43)$$

The two terms are orthogonal because $P_R P_0 = 0$, so

$$\|e\|_2^2 = \langle P_R e + P_0 e, P_R e + P_0 e \rangle \quad (44)$$

$$= \|P_R e\|_2^2 + \|P_0 e\|_2^2 + 2 \langle P_R e, P_0 e \rangle \quad (45)$$

$$= \|P_R e\|_2^2 + \|P_0 e\|_2^2. \quad (46)$$

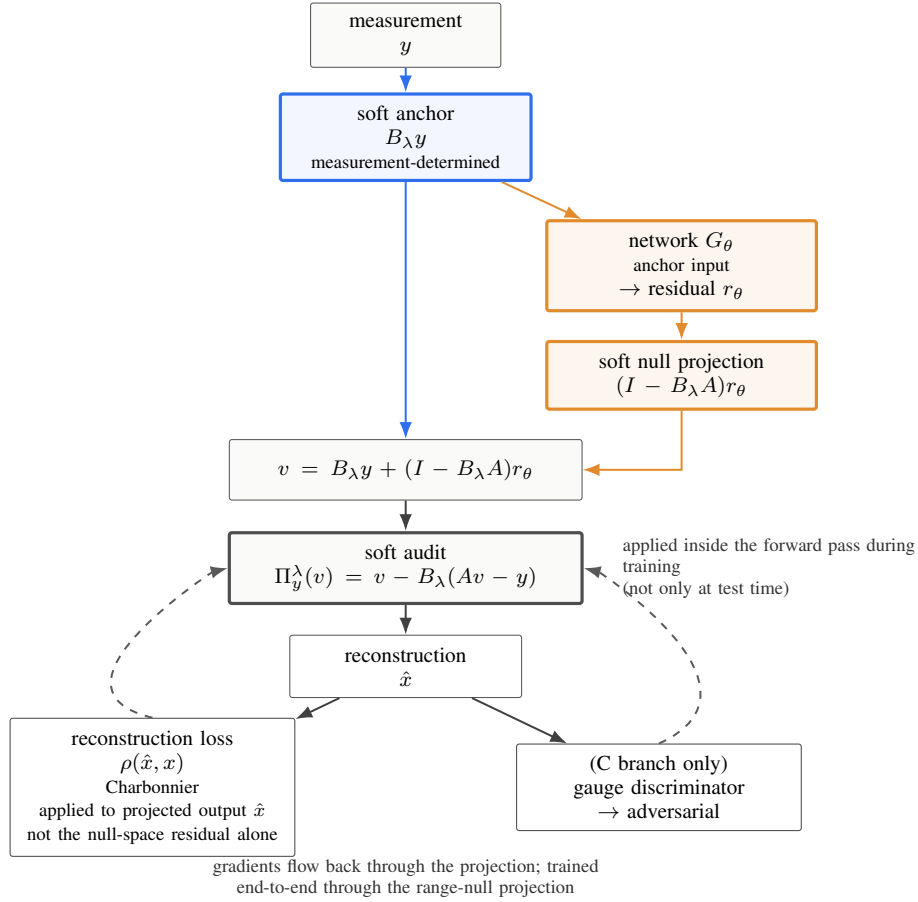


Fig. 2. Simplified training pipeline. The measurement gives a soft anchor $B_\lambda y$, the network supplies a null-space completion through the soft null projection $I - B_\lambda A$, and the soft audit Π_y^λ is applied inside the training forward pass before the reconstruction loss. The reconstruction loss is applied to the projected full image $\hat{x}$, not to the null-space residual alone; thus gradients pass through the range-null projection while the C branch adds a gauge adversarial term. The blue path marks measurement-determined row-space content and the orange path marks prior-supplied null-space content. The exact geometric decomposition uses $P_R = A^\dagger A$ and $P_0 = I - A^\dagger A$, whereas the implementation uses the noise-aware soft operators $B_\lambda = A^\top(AA^\top + \lambda I)^{-1}$ and Π_y^λ with $\lambda > 0$ (the code uses $\lambda = 0.001$). For clarity this shows the conceptual data flow; the full implementation is a two-stage reconstructor with a refinement stage and two soft-audit applications, simplified here as one audit block, and uses the soft (ridge) projection B_λ with $\lambda > 0$, which approaches the exact projection as $\lambda \rightarrow 0$.

Define the pre-audit row-error share

$$s = \frac{\|P_R e\|_2^2}{\|e\|_2^2}. \quad (47)$$

Then

$$\|P_0 e\|_2^2 = \|e\|_2^2 - \|P_R e\|_2^2 \quad (48)$$

$$= (1 - s)\|e\|_2^2. \quad (49)$$

If an ideal hard audit removes all row-space error and leaves $P_0 e$ unchanged, the pre-audit and post-audit mean-squared errors satisfy

$$\frac{\text{MSE}_{\text{post}}}{\text{MSE}_{\text{pre}}} = \frac{\|P_0 e\|_2^2/n}{\|e\|_2^2/n} = 1 - s. \quad (50)$$

For a fixed image dynamic range, the maximum achievable PSNR gain, attained in the limit $\lambda \rightarrow 0$ by exact range-space

projection, is

$$\Delta\text{PSNR}_{\text{max}} = 10 \log_{10} \left(\frac{\text{MSE}_{\text{pre}}}{\text{MSE}_{\text{post}}} \right) \quad (51)$$

$$= 10 \log_{10} \left(\frac{1}{1 - s} \right) \quad (52)$$

$$= -10 \log_{10}(1 - s). \quad (53)$$

For the actual soft audit with $\lambda > 0$, Section III shows that each measured singular mode is multiplied by $\lambda/(\lambda + \sigma_i^2)$. Therefore the row-space error is contracted rather than removed, and the realized PSNR gain is a contracted version of (53), strictly below $\Delta\text{PSNR}_{\text{max}}$ whenever any visible error remains after the soft audit. This upper bound explains the observed split: trained networks have a small pre-audit row-error share, so the bound is near zero and the audit barely moves PSNR; a reconstruction with large pre-audit row-error share would have large range-space PSNR headroom. This version restricts the quantitative PSNR claim to the reported rows.

TABLE I
POST-HOC MEASUREMENT AUDIT ON DIFFERENT RECONSTRUCTOR FAMILIES. THE SCR-5 AUDIT ROWS ARE SEPARATE FROM THE GAN COMPARISON METRICS IN TABLE XIII. CS-TV IS SMALL-SUBSET EVIDENCE, NOT A FULL-SCALE BASELINE CLAIM.

Family	Regime	N	PSNR before	PSNR after	ΔPSNR	RelMeasErr before $\rightarrow$ after
BP	Rad-5	2000	7.297	7.297	0.0000	$5.17 \times 10^{-5} \rightarrow 3.04 \times 10^{-9}$
BP	Scr-5	2000	14.310	14.310	0.0000	$1.05 \times 10^{-7} \rightarrow 1.05 \times 10^{-10}$
Tikhonov	Rad-5	2000	6.667	6.667	2.21×10^{-10}	$5.18 \times 10^{-5} \rightarrow 3.05 \times 10^{-9}$
Tikhonov	Scr-5	2000	14.310	14.310	2.77×10^{-5}	$9.99 \times 10^{-4} \rightarrow 9.98 \times 10^{-7}$
CS-TV	Rad-5	8	8.872	8.872	7.12×10^{-7}	$1.11 \times 10^{-3} \rightarrow 6.38 \times 10^{-8}$
CS-TV	Scr-5	8	13.190	13.190	2.54×10^{-4}	$3.20 \times 10^{-3} \rightarrow 3.19 \times 10^{-6}$
Learned	Rad-5	2000	22.192	22.206	0.0136	$3.68 \times 10^{-2} \rightarrow 1.90 \times 10^{-6}$
Learned	Scr-5	2000	22.146	22.185	0.0387	$1.80 \times 10^{-2} \rightarrow 1.80 \times 10^{-5}$

TABLE II
RANGE-SHARE LAW UNDER THE PER-IMAGE MEAN-REMOVED CONVENTION. UNDER THIS CONVENTION THE ROW-SPACE ENERGY SHARE TRACKS THE SAMPLING RATE FOR BOTH RADEMACHER AND SCRAMBLED HADAMARD SENSING.

Regime	ρ mean	Ceiling	BP PSNR
Rad-5	0.050	14.304	7.297
Scr-5	0.052	14.311	14.310
Rad-10	0.101	14.541	7.756
Scr-10	0.099	14.534	14.533

Plain-language interpretation: auditing can only buy image-quality improvement from the part of the pre-audit error that lies in the measurement-visible row space; a soft audit buys only the contracted part of that improvement. If most remaining error is null-space prior content, PSNR and measurement residual become almost decoupled.

B. Range Share and Observed Flatness

The experiments include a row-space share table for the measured component of the image. Table II reports these values. Its ρ is a row-space energy share of the image under the stated image convention, not the pre-audit row-error share s in (47). The two quantities use the same range-null orthogonality but answer different questions: ρ explains row-space anchor ceilings such as BP, while s controls the maximum PSNR gain available to a particular pre-audit reconstruction. The table is therefore not used as a numerical estimate of trained-network pre-audit s . The main table uses the per-image mean-removed convention so that DC handling does not dominate the reported row-space share.

The raw-convention rows are retained only for provenance.¹ The gap between Rad-5 BP and its ceiling reflects the DC/global-mean coverage difference under Rademacher sensing noted in the footnote, whereas scrambled Hadamard sensing preserves the DC component more directly and reaches the corresponding BP ceiling.

¹The raw-convention rows are Rad-5 $\rho = 0.050$, Scr-5 $\rho = 0.804$, Rad-10 $\rho = 0.102$, and Scr-10 $\rho = 0.814$. They are not used in Table II because the DC/mean component is handled differently under that convention.

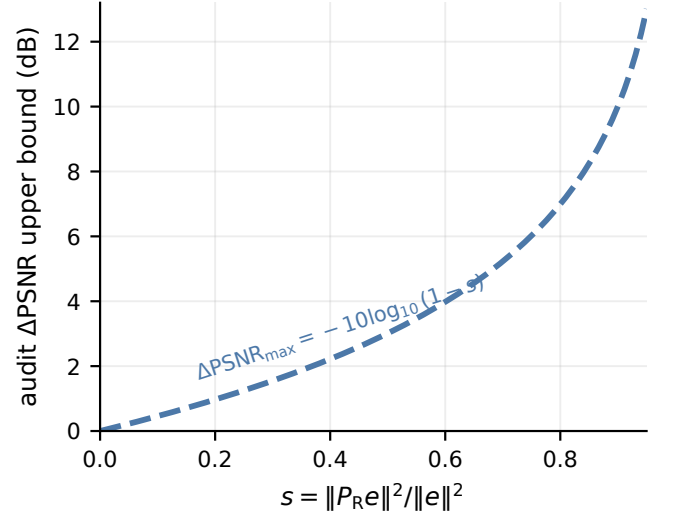


Fig. 3. Theorem 1 hard-projection PSNR upper bound, $\Delta\text{PSNR}_{\max} = -10\log_{10}(1 - s)$, shown as a function of the pre-audit row-space error share s . The measured flat-corner behavior of trained networks is reported numerically in Tables I and III: the audit moves PSNR by only 0.0136 dB for Rad-5 and 0.0387 dB for Scr-5 while reducing measurement residuals by orders of magnitude.

For learned reconstructions, the post-hoc rows in Table I show the flat-corner behavior: Rad-5 learned output moves from 22.192 to 22.206 dB while RelMeasErr moves from 3.68×10^{-2} to 1.90×10^{-6} ; Scr-5 learned output in the main audit experiment moves from 22.146 to 22.185 dB while RelMeasErr moves from 1.80×10^{-2} to 1.80×10^{-5} . Measurement accountability changes by orders of magnitude while PSNR changes by only 0.0136 dB and 0.0387 dB.

C. Ablation Evidence

Table III reports the 5% ablation rows. Removing the final audit or measurement loss leaves PSNR near the full learned output, but the pre-audit measurement residual is much worse. Applying the audit restores the residual with little PSNR movement. This is the empirical expression of (46): image quality and measurement accountability are coupled only through the row-space error share.

TABLE III

ABLATION AND POST-HOC AUDIT ROWS. PSNR CHANGES ARE SMALL, WHILE RelMeasErr CHANGES BY ORDERS OF MAGNITUDE, SUPPORTING THE GEOMETRIC SEPARATION BETWEEN IMAGE QUALITY AND MEASUREMENT ACCOUNTABILITY.

Row	PSNR pre	PSNR post	Δ PSNR	RelMeasErr pre	RelMeasErr post	contraction
rad5_no_final_audit	22.192	22.206	0.0136	3.68×10^{-2}	1.90×10^{-6}	5.15×10^{-5}
rad5_no_gate_no_final_audit	22.186	22.200	0.0136	3.69×10^{-2}	1.90×10^{-6}	5.15×10^{-5}
rad5_no_final_audit_no_meas_loss	22.180	22.219	0.0384	7.30×10^{-2}	3.76×10^{-6}	5.15×10^{-5}
scr5_no_final_audit	22.146	22.185	0.0387	1.80×10^{-2}	1.80×10^{-5}	9.99×10^{-4}
scr5_no_gate_no_final_audit	22.146	22.185	0.0385	1.80×10^{-2}	1.80×10^{-5}	9.99×10^{-4}
scr5_no_final_audit_no_meas_loss	22.138	22.171	0.0322	2.09×10^{-2}	2.09×10^{-5}	9.99×10^{-4}

The Scr-5 ablation rows in Table III are used only as audit evidence; the Scr-5 GAN comparison metrics are reported separately in Table XIII.

V. WHAT THE CERTIFICATE CANNOT DO: THE VERIFIABILITY BARRIER

The contraction certificate proves a statement about the measured component of an image. It does not prove that the unmeasured component is the one that would have appeared in the scene. This limitation is not a weakness of the particular audit in Section III; it follows from the same range-null decomposition that made the audit possible.

A. A Feasible Wrong Image

Let x_i and x_j be two images, and suppose that we want an image that has the measurement of x_i while retaining the null-space content of x_j . Define

$$u_{ij} = x_j - A^\dagger(Ax_j - y_i), \quad y_i = Ax_i. \quad (54)$$

We now compute its measured component explicitly. Since $y_i \in \text{range}(A)$ and $Ax_j \in \text{range}(A)$, their difference $Ax_j - y_i$ lies in $\text{range}(A)$. The matrix $AA^\dagger$ is the orthogonal projector onto $\text{range}(A)$, so

$$Au_{ij} = Ax_j - AA^\dagger(Ax_j - y_i) \quad (55)$$

$$= Ax_j - (Ax_j - y_i) \quad (56)$$

$$= y_i. \quad (57)$$

Thus u_{ij} is perfectly measurement-consistent with the measurement of x_i . Its null-space component is

$$P_0 u_{ij} = P_0 x_j - P_0 A^\dagger(Ax_j - y_i) \quad (58)$$

$$= P_0 x_j, \quad (59)$$

The second equality holds because $P_0 A^\dagger = 0$: $A^\dagger$ maps into the row space and P_0 annihilates that space. Its row-space component is fixed by y_i :

$$P_R u_{ij} = A^\dagger A u_{ij} = A^\dagger y_i = P_R x_i. \quad (60)$$

In plain language, one can splice the measured part of one image to the unmeasured part of another image and obtain a reconstruction that passes the measurement audit exactly.

Table IV reports the Rad-5 feasible-pair construction. The constructed wrong images have residuals at floating-point scale while remaining visually wrong with respect to the target image. These rows are the cleanest empirical reminder that consistency is not truth.

TABLE IV

FEASIBLE WRONG-IMAGE CONSTRUCTION FOR RAD-5. THE CONSTRUCTED IMAGE u_{ij} IS MEASUREMENT-CONSISTENT WITH y_i BUT KEEPS $P_0 x_j$. THEREFORE A SMALL RESIDUAL CERTIFIES THE MEASURED COMPONENT, NOT SEMANTIC CORRECTNESS.

Family	pairs	RelMeasErr(u_{ij}, y_i)	truth residual	PSNR(u_{ij}, x_i)
Rad-5	8	$2.16 \times 10^{-15} - 4.00 \times 10^{-15}$	$3.45 \times 10^{-3} - 7.46 \times 10^{-3}$	7.697–11.378

Rad-5 pair $i = 1789, j = 935$: wrong image matches y_i more tightly than truth

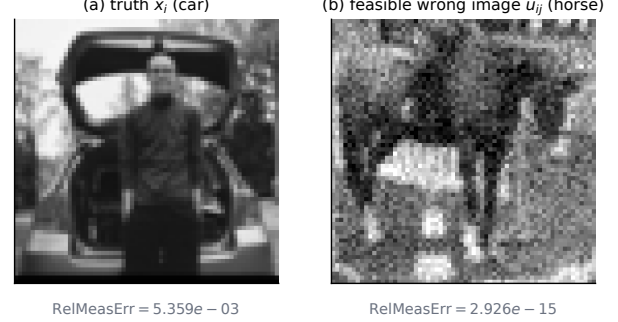


Fig. 4. Feasible hallucination for a Rad-5 feasible-pair output ($i = 1789, j = 935$). The true target image is a car with $\text{RelMeasErr}(x_i, y_i) = 5.36 \times 10^{-3}$, while the constructed wrong image $u_{ij} = x_j - A^\dagger(Ax_j - y_i)$ visually follows the horse donor and has $\text{RelMeasErr}(u_{ij}, y_i) = 2.93 \times 10^{-15}$. Thus an image can satisfy the measurement record more tightly than the noisy truth while remaining semantically wrong.

B. Magnitude Is Not Pixelwise Correctness

The previous construction shows that an audit cannot identify the correct null-space image. A second, more operational question is whether the observable map $|P_0 \hat{x}|$ can be used as a pixelwise error map. We call it a prior-supplied-content map: it shows where the prior supplied null-space content, not where the reconstruction is wrong. The range-null decomposition again gives the relevant algebra. Let $\hat{x}$ be any audited reconstruction. Then the null-space error is

$$P_0(\hat{x} - x) = P_0 \hat{x} - P_0 x. \quad (61)$$

The observable magnitude $|P_0 \hat{x}|$ contains no direct term for the unknown $P_0 x$; it is a property of the reconstruction, not an error certificate. A large value in the prior-supplied-content map may be accurate texture, inaccurate texture, or a harmless choice among feasible completions. Magnitude does not indicate correctness.

TABLE V

THE PRIOR-SUPPLIED-CONTENT MAP $|P_0\hat{x}|$ IS NOT A PIXELWISE ERROR MAP ON THE 96×96 RAD-5 VALIDATION. SPEARMAN AND PEARSON CORRELATIONS ARE NEAR ZERO. FOR ARMS B AND C, THE TOP-10% $|P_0\hat{x}|$ PIXELS HAVE LOWER ACTUAL P_0 ERROR THAN THE REMAINING PIXELS.

Arm	Spearman	Pearson	top-10% error	rest-90% error
A	-0.079	-0.069	0.259	0.242
B	0.073	0.061	0.071	0.079
C	0.072	0.061	0.071	0.079

Table V reports the 96×96 Rad-5 validation. For the learned arms B and C, the pixels with the largest $|P_0\hat{x}|$ do not have larger actual null-space error; the top-10% error is lower than the remaining 90%. The negative result is the point: null-space magnitude and null-space correctness are empirically decoupled.

C. Amount, Distribution, and Correctness Are Separate

Posterior sampling makes the boundary sharper. For a fixed y , a sampler returns $\hat{x}^{(k)} = \Pi_y^\lambda(v^{(k)})$. By Section III, $A\hat{x}^{(k)} \approx y$ for every k when λ is small. However, the sample variance splits as

$$\text{Var}(\hat{x}^{(k)}) = \text{Var}(P_R\hat{x}^{(k)} + P_0\hat{x}^{(k)}) \quad (62)$$

$$= \text{Var}(P_R\hat{x}^{(k)}) + \text{Var}(P_0\hat{x}^{(k)}), \quad (63)$$

where the cross term vanishes because P_R and P_0 project onto orthogonal subspaces. The audit can drive the first term to the measurement noise floor, but it says neither how large the second term should be nor whether its mean is centered on the true P_0x . Thus the amount of uncertainty, the distribution of samples, and the correctness of the unmeasured content are three different questions.

Table VI summarizes the posterior-calibration diagnostics, with the full gate-by-gate table in Appendix A. A deterministic Rad-5 checkpoint collapses at the 10^{-3} pixel-std scale and 10^{-6} P_0 -variance scale. Replacing full reconstruction loss with row-space-only reconstruction restores null-space variation to the 10^{-2} pixel-std scale and 10^{-3} P_0 -variance scale. The restored variation is not merely white noise: the spectral sanity gate passes a non-white, low-frequency-structured check. The standard deviations, variances, slope, and high-to-low ratio are summarized in Table VI and reported gate-by-gate in Table XIV. In plain language, the collapse was not an architectural dead end; the full reconstruction loss was suppressing feasible null-space diversity, and removing that pressure lets structured null variation appear.

Calibration remains unresolved for a different reason. The uncentered calibration audit has subnominal 90% coverage and a nonzero P_0 mean offset: pairwise diversity can increase spread while leaving the sample cloud centered in the wrong place. Adding a P_0 mean anchor improves the center but collapses the spread; the best anchor/diversity scan still reaches only about 45% pixel coverage and 48% P_0 coverage at the nominal 90% level. The results identify the base bottleneck: the deterministic base estimate has P_0 det-to-GT RMSE about

0.081, while the best scan has P_0 mean-to-GT RMSE about 0.088. Exact coverage, offset, and RMSE values are summarized in Table VI and fully listed in Table XIV. Plainly, a sampler can learn to move inside the null-space affine set, but if the base reconstruction is centered on the wrong P_0 content, spread tuning alone cannot make the posterior calibrated.

The gate-by-gate coverage diagnostics are reported in Table XIV; no separate coverage-evolution figure is needed for this evidence.

D. Operator Drift Is Visible to the Certificate

The measurement certificate is also useful when the deployed operator is not the operator assumed by the audit. We include a simulation-scoped operator-drift experiment: the audit is performed with a drifted operator, and the residual is then evaluated against the original true operator. This is not a hardware-robustness claim. It is a calibration-mismatch probe inside the simulation setup.

Table VII shows the pattern. For Rad-5, increasing the relative operator drift from 0 to 0.05 changes post-audit PSNR only from 22.206 dB to 22.178 dB, while the residual against the true operator rises from 1.90×10^{-6} to 4.88×10^{-2} . For Scr-5, the same drift range changes PSNR from 22.185 dB to 22.155 dB, while the true-operator residual rises from 1.80×10^{-5} to 1.26×10^{-2} . In plain language, PSNR barely sees the calibration mismatch, while the certificate does.

E. Wrong Measurements Break the Reconstruction

A separate dependence test asks whether the learned reconstructor uses the recorded measurement or merely emits plausible samples from the data prior. The experiment perturbs the measurement input in two ways: by rolling the batch so that each image receives another image's y , and by shuffling the measurement coordinates. Table VIII reports 500-image probes. Across Rad-5, Scr-5, Rad-10, and Scr-10, wrong- y inputs reduce PSNR by 12.174–14.793 dB, and shuffled- y inputs reduce PSNR by 14.537–17.026 dB. In plain language, the model is not acting as a pure dataset prior; it depends strongly on the recorded measurement.

This dependence does not contradict the range-null geometry in Section IV. The linear operator A geometrically constrains only the row-space component: every feasible image may still differ by an arbitrary P_0 component. A trained conditional reconstructor is a different object: it uses y as an input to choose which prior-supplied P_0 completion to emit. Thus the wrong- y collapse shows that the network really conditions its null-space completion on the recorded measurement; it does not mean that the measurement geometry itself constrains the null space.

F. The Unified Boundary

The three verifiability-barrier tests in Sections V-A–V-C are one theorem-shaped message. Equation (59) shows that null-space correctness cannot be certified by $A\hat{x} = y$. Equation (61) shows that null-space magnitude is not an error map because the unknown term P_0x remains absent. Equation (63) shows

TABLE VI

POSTERIOR-SAMPLING DIAGNOSTIC SUMMARY. THE ROW-SPACE-ONLY DIAGNOSTIC RESTORES P_0 VARIATION WHILE PRESERVING MEASUREMENT CONSISTENCY, BUT THE ANCHOR/DIVERSITY SCAN REMAINS UNDER-CALIBRATED AND THE BASE P_0 ESTIMATE IS OFFSET FROM GROUND TRUTH. FULL GATE-BY-GATE DIAGNOSTICS ARE IN TABLE XIV.

Gate	diagnostic	fixed- y std	P_0 var	max RelMeasErr	key posterior evidence
1	deterministic Rad-5 baseline	0.0010	1.09×10^{-6}	3.84×10^{-5}	collapse
2	row-space-only reconstruction	0.0335	1.28×10^{-3}	5.61×10^{-5}	anti-collapse pass
6	anchor/diversity scan	0.0349	1.39×10^{-3}	5.62×10^{-5}	90% pixel/ P_0 cov. 0.453 / 0.478; offset 0.0331
7	base P_0 bottleneck	n.r.	n.r.	n.r.	det-to-GT RMSE 0.0810; best mean-to-GT RMSE 0.0877

TABLE VII

SIMULATION-SCOPED OPERATOR-DRIFT PROBE. THE AUDIT USES A DRIFTED OPERATOR, WHILE RelMeasErr is EVALUATED AGAINST THE TRUE OPERATOR. PSNR IS NEARLY FLAT, BUT THE CERTIFICATE RESIDUAL RISES BY ORDERS OF MAGNITUDE.

Regime	drift	post RelMeasErr	true A	post PSNR
Rad-5	0.000	1.90×10^{-6}		22.206
Rad-5	0.005	4.89×10^{-3}		22.206
Rad-5	0.010	9.79×10^{-3}		22.205
Rad-5	0.020	1.96×10^{-2}		22.201
Rad-5	0.050	4.88×10^{-2}		22.178
Scr-5	0.000	1.80×10^{-5}		22.185
Scr-5	0.005	1.26×10^{-3}		22.184
Scr-5	0.010	2.53×10^{-3}		22.183
Scr-5	0.020	5.06×10^{-3}		22.180
Scr-5	0.050	1.26×10^{-2}		22.155

TABLE VIII

WRONG-MEASUREMENT DEPENDENCE TEST. REPLACING THE MEASUREMENT RECORD BY ANOTHER SAMPLE'S y OR SHUFFLING MEASUREMENT COORDINATES CAUSES LARGE PSNR DROPS, SHOWING THAT THE RECONSTRUCTOR USES THE RECORDED MEASUREMENT.

Model	clean	wrong- y	drop	shuffled- y	drop
main_rad5	22.202	9.962	12.241	7.481	14.721
main_scr5	22.160	9.986	12.174	7.624	14.537
main_rad10	24.678	9.886	14.793	7.652	17.026
main_scr10	24.604	9.887	14.717	7.597	17.007

that sample diversity can be created in P_0 while its center and coverage remain wrong. In plain language, the amount of unmeasured content, the distribution of unmeasured content, and the correctness of unmeasured content are decoupled.

This is the verifiability barrier. The certificate guarantees row-space accountability: the reconstruction agrees with the measurement-supported component up to the contraction/noise floor. It cannot verify the correctness of prior-supplied P_0 content, and the experiments show that two tempting workarounds also fail: $|P_0 \hat{x}|$ does not localize error, and learned measurement-consistent sampling does not automatically calibrate the posterior. These are not separate disappointments. Section V-B is the single-reconstruction face of the barrier: after seeing one audited image, the magnitude of $P_0 \hat{x}$ does not reveal where $P_0 \hat{x} - P_0 x$ is large. Section V-C is the sampling face of the same barrier: even a cloud of measurement-consistent samples remains limited by the base estimator's P_0 bias, because the measurement provides no

TABLE IX

GAUGE DIAGNOSTICS AND REGIME GATING. THE GAN BRANCH IS USED ONLY WHERE THE DIAGNOSTIC SIGNAL IS PRESENT; WEAK-GATE 10% REGIMES ARE STOPPED RATHER THAN USED FOR A QUALITY CLAIM.

Regime/diagnostic	AUC	CI/source	decision
Scr-5 patch-GAN unconditional gauge	0.847	0.824–0.874	positive
Scr-5 conditional image-data gauge	0.731	0.696–0.758	positive
Scr-5 residual-feature logistic	0.980	0.967–0.990	positive
Rad-5 regime map	0.877	0.845–0.907	train/evidence positive
Scr-10 regime map	0.624	0.579–0.670	weak gate; stop
Rad-10 regime map	0.640	0.590–0.677	weak gate; stop

corrective signal for that bias. The obstruction in both cases is exactly $AP_0 = 0$.

VI. A GENERATIVE PRIOR UNDER THE CERTIFICATE: GAN AS A CASE STUDY

We use a GAN branch as a controlled example of a learned prior operating inside the range-null geometry. The GAN is not the main claim of this paper, and we do not claim state-of-the-art perceptual quality or superiority over diffusion models. The purpose is narrower: to show how a generative prior can change audited null-space content while the measurement certificate remains attached to the output.

A. Gauge-Equalized Discrimination

An unconditional discriminator can learn shortcuts in the measured component. A range-null reading makes this failure mode explicit. If the discriminator sees $P_R \hat{x}$ differences caused by measurement residual, it may reward or punish measurement artifacts instead of image prior structure. The gauge-equalized branch is designed to reduce this shortcut by focusing the adversarial signal on content that is not already certified by the measurement audit.

The diagnostic results support the presence of such signal. Table IX summarizes the Scr-5 gauge AUCs and the regime map: the patch-GAN unconditional gauge is about 0.85 AUC, the conditional image-data gauge is about 0.73 AUC, and the residual-feature logistic diagnostic is about 0.98 AUC. Scr-5 and Rad-5 pass the train/evidence gate, while Scr-10 and Rad-10 are weak-gate regimes where the GAN branch is not advanced.

The shortcut stress test in Table X shows why the distinction matters. At $\alpha = 0.1$, row-space perturbations produce a standard discriminator fake-score delta of about 0.477, whereas the gauge-discriminator row delta is 0. Null-space perturbations produce a gauge-discriminator delta of about 0.739. This is

TABLE X

SHORTCUT STRESS TEST AT $\alpha = 0.1$. THE GAUGE DISCRIMINATOR IS INSENSITIVE TO ROW-SPACE PERTURBATIONS AND RESPONDS TO NULL-SPACE PERTURBATIONS, WHICH ALIGNS THE PRIOR SIGNAL WITH THE UNMEASURED COMPONENT.

Model	base	perturbation	delta	mean RelMeasErr
standard D	real	row	0.384	1.13×10^{-1}
standard D	real	null	0.401	5.55×10^{-3}
standard D	fake mean	row	0.477	1.12×10^{-1}
standard D	fake mean	null	0.485	5.58×10^{-3}
gauge D	real	row	0.000	1.13×10^{-1}
gauge D	real	null	0.540	5.55×10^{-3}
gauge D	fake mean	row	0.000	1.12×10^{-1}
gauge D	fake mean	null	0.739	5.58×10^{-3}

TABLE XI

96 $\times$ 96 RAD-5 CONTROLLED GAN, TEN PAIRED SEEDS. C-B IS NEGATIVE FOR LPIPS AND RAPSD IN ALL TEN SEEDS, WHILE C-B PSNR IS A REAL BUT TINY POSITIVE SHIFT UNDER THE SOFT AUDIT. RAD-5 RelMeasErr STAYS ON THE AUDITED 10^{-4} SCALE.

Quantity	sign consistency	mean C-B / value	95% CI or support
LPIPS	10/10 negative	-0.00552	per-sample bootstrap excludes 0
RAPSD	10/10 negative	-7.48×10^{-4}	per-sample bootstrap excludes 0
PSNR	10/10 positive	+0.0148 dB	t : [+0.0107, +0.0189] dB; boot: [+0.0113, +0.0180] dB
RelMeasErr	-	B: 9.28×10^{-5} ; C: 9.90×10^{-5}	B/C per-sample max: 1.61×10^{-5}

the desired direction for a prior module: the adversarial signal should not be a surrogate measurement-residual detector.

B. Controlled 96 $\times$ 96 Rad-5 Result

Table XI reports the controlled 96 $\times$ 96 Rad-5 GAN paired-seed comparison over ten seeds. The gauge branch C improves LPIPS and RAPSD relative to the baseline branch B with the same sign in all ten paired seeds: mean C-B LPIPS is -0.00552 and mean C-B RAPSD is -7.48×10^{-4} . The within-seed per-sample bootstrap intervals exclude zero, and the ten-seed sign consistency is substantially stronger than the earlier three-seed probe.

The ten-seed PSNR result is also sign-consistent, but its interpretation is different. C-B PSNR is positive in 10/10 seeds with seed-level mean +0.0148 dB, a t 95% confidence interval of [+0.0107, +0.0189] dB, and a bootstrap 95% confidence interval of [+0.0113, +0.0180] dB. Because B and C are identical except for the gauge adversarial branch, that branch is the source of this displacement. The observation is consistent with the soft-audit framework: under a hard projection ($\lambda \rightarrow 0$), the row-space component of both branches would be pinned by the measurement, whereas the soft audit ($\lambda > 0$) leaves a small margin in which the branches can differ slightly. We do not localize this gain to a specific component, row-space or null-space. The gain is at the 0.01–0.02 dB level and is not a quality claim; the perceptual/spectral improvements and the unchanged certificate remain the point. B and C remain measurement accountable: the seed-mean RelMeasErr averages are 9.28×10^{-5} and 9.90×10^{-5} , respectively, and the B/C per-sample maximum is 1.61×10^{-4} .

C. Standard cGAN Control

A fair reading also requires a standard cGAN control. Table XII shows the 96 $\times$ 96 Rad-5 standard-control focus

metrics. The standard branch D and the gauge branch C are comparable and mixed: the seed-level wins go in both directions and the magnitudes are small. This prevents the GAN section from becoming a dominance claim.

The small effect size is also geometrically expected. For two audited reconstructions of the same measurement, $\hat{x}_1$ and $\hat{x}_2$, hard consistency would give $P_R \hat{x}_1 = P_R \hat{x}_2 = A^\dagger y$, and soft consistency makes this equality approximate. Their difference is therefore mostly null-space:

$$\hat{x}_1 - \hat{x}_2 = P_R(\hat{x}_1 - \hat{x}_2) + P_0(\hat{x}_1 - \hat{x}_2) \approx P_0(\hat{x}_1 - \hat{x}_2). \quad (64)$$

For a fixed ground truth x , write $e_i = \hat{x}_i - x$. The squared error changes as

$$\|e_2\|_2^2 - \|e_1\|_2^2 = \|P_R e_2\|_2^2 - \|P_R e_1\|_2^2 \quad (65)$$

$$+ \|P_0 e_2\|_2^2 - \|P_0 e_1\|_2^2. \quad (66)$$

The first difference is small after auditing, so any perceptual improvement from the GAN prior must come mostly from changing P_0 content. Plainly, the prior can make the null-space content look more natural without guaranteeing a lower Euclidean distance to the hidden true $P_0 x$; that is why LPIPS/RAPSD can move while PSNR barely moves.

D. Controlled Scr-5 Metrics

Because several Scr-5 training protocols were evaluated during development, Table XIII is the only table in this section that reports Scr-5 PSNR, SSIM, LPIPS, RAPSD, and RelMeasErr values for the GAN comparison. The section does not mix Scr-5 metrics across protocol variants.

VII. RELATED WORK

A. Range-Null Diffusion and DDNM-Style Solvers

Range-null diffusion and DDNM-style inverse solvers also use a decomposition of measured and unmeasured components [CITE]. The delta here is the certification question: we use the decomposition to define a test-time measurement audit, to prove its contraction certificate, and to state what the certificate cannot verify. The GAN branch also introduces a discriminator-shortcut issue that is orthogonal to diffusion sampling: an adversarial discriminator can learn row residual shortcuts unless its input is gauge-equalized.

B. Adversarial Regularization

Adversarial regularizers have long been used to make inverse-problem outputs look more natural [CITE]. Our distinction is that the prior is placed under a measurement certificate. The discriminator is not treated as a quality oracle: it is audited for shortcut sensitivity, and its effect is interpreted as prior-supplied P_0 content rather than certified truth.

C. Deep Null-Space Learning

Deep null-space learning explicitly trains networks to fill the null component of an inverse problem [CITE]. In the notation of this paper, the supervised null-space arm B is a representative of that idea. We add a soft, noise-aware

TABLE XII

96 × 96 RAD-5 STANDARD CGAN CONTROL. D–C DIFFERENCES ARE MIXED ACROSS SEEDS AND METRICS, SUPPORTING THE LIMITED CLAIM THAT THE GAUGE BRANCH IS A CONTROLLED PRIOR EXAMPLE RATHER THAN A UNIVERSAL IMPROVEMENT OVER A STANDARD CGAN.

Seed	metric	C	D_standard	D–C
1	PSNR	19.789	19.788	-0.0017
1	SSIM	0.458	0.458	-0.0001
1	RAPSD	0.0089	0.0089	0.0000
1	LPIPS	0.411	0.411	0.0002
2	PSNR	19.788	19.788	0.0008
2	SSIM	0.458	0.459	0.0004
2	RAPSD	0.0089	0.0089	0.0000
2	LPIPS	0.415	0.415	0.0000
3	PSNR	19.788	19.788	-0.0004
3	SSIM	0.459	0.459	0.0000
3	RAPSD	0.0087	0.0087	0.0000
3	LPIPS	0.413	0.413	0.0000

TABLE XIII

CONTROLLED SCR-5 5% METRICS. THESE ARE THE ONLY SCR-5 GAN-BRANCH QUALITY/ACCOUNTABILITY METRICS USED IN SECTION VI. FOR THIS SCR-5 REGIME THE AUDITED RelMeasErr VALUES ARE ON THE 10^{-3} SCALE, UNLIKE THE RAD-5 96 × 96 ROWS IN TABLE XI.

Arm	PSNR	SSIM	LPIPS	RAPSD	RelMeasErr
B	22.262	0.627	0.2349	0.0054	5.56×10^{-3}
C_gauge	22.257	0.628	0.2308	0.0052	5.57×10^{-3}
D_standard	22.258	0.628	0.2310	0.0052	5.57×10^{-3}

certificate Π_y^λ , and we analyze the verifiability of the learned P_0 content itself. The negative map and posterior-calibration results show that producing a plausible null-space estimate is different from certifying where it is wrong or calibrating its distribution.

D. Data Consistency and Projection Layers

Projection and data-consistency layers enforce agreement with the measurement operator in many reconstruction networks [CITE]. Our audit $\Pi_y^\lambda(v) = v - B_\lambda(Av - y)$ can be read as a soft data-consistency layer, but the contribution is semantic and geometric: we derive its singular-mode contraction, separate its PSNR effect from its accountability effect, and state the feasible-hallucination boundary that no projection layer can cross.

E. Diffusion Inverse Solvers

DPS, DDRM, IIGDM, and related diffusion inverse solvers are important quality baselines [CITE]. We do not claim to beat them. Our claim is about certification and verifiability rather than state-of-the-art reconstruction: even a stronger sampler can only be certified in P_R by the measurement, while its P_0 center, spread, and correctness must be evaluated separately. In plain language, better generative modeling may improve the prior-supplied content, but it does not erase the range-null boundary.

VIII. DISCUSSION AND CONCLUSION

The main message of this paper is that under-sampled ghost imaging has a two-part geometry. The row-space component $P_R x = A^\dagger A x$ is the part the measurement can see, and the null-space component $P_0 x = (I - A^\dagger A)x$ is the part the

measurement cannot see. Once this split is made explicit, several observations that can look empirical become the same mathematical fact. The audit in Section III contracts measurement residuals because it acts on the row-space residual. The PSNR response in Section IV is small for trained networks because their remaining error is mostly outside the row-space share that the audit can change. The feasible hallucinations and posterior-calibration failures in Section V persist because the measurement cannot identify the correct $P_0 x$.

The practical recommendation is therefore not to replace image-quality metrics with measurement residuals, nor to replace measurement residuals with image metrics. They answer different questions. PSNR, SSIM, LPIPS, RAPSD, and related metrics evaluate agreement with a reference or perceptual proxy when such a reference is available. The audit evaluates whether the reported image is accountable to the measurement available at deployment time. A trustworthy pipeline should report both, and should state clearly which part of the image is measured, which part is prior-supplied, and which claims cannot be verified without ground truth.

A natural concern is whether enforcing measurement consistency degrades the naturalness of the reconstruction relative to an unconstrained end-to-end network. In our pipeline this is not the case for a structural reason: the reconstruction loss is applied to the final projected output $\hat{x} = \Pi_y^\lambda(\cdot)$ rather than to the null-space residual in isolation, and the projection is part of the forward pass during training. The network is therefore trained end-to-end through the range-null projection, learning to produce a natural full image under the constraint that its row-space component is locked to $A^\dagger y$. Empirically the cost of the projection is negligible: the audit moves PSNR by less than 0.04 dB (Tables I and III) while the audited reconstructions retain strong perceptual quality (LPIPS/RAPSD, Section VI).

Naturalness is supplied by the trained prior and measurement consistency is guaranteed by the audit; because the prior is trained under the audit, the two are obtained jointly rather than traded off. We emphasize, however, that the audit itself does not enforce naturalness: a measurement-consistent reconstruction can be unnatural, as in the constructed u_{ij} in Fig. 1, which is precisely the measurement-consistency $\neq$ correctness $\neq$ naturalness distinction.

There are important limitations. The experiments in this manuscript are simulation-based; hardware ghost-imaging validation is left to future work. The posterior-sampling line shows restored null-space diversity but not calibrated coverage. The GAN branch gives a controlled example of a prior under audit, not a claim that GANs are the best generative model for this problem; its effect is small and comparable to a standard cGAN control. The experimental scope is one ghost-imaging problem. The range-null principle applies to general linear inverse problems, but cross-domain validation is outside the scope of this study; any inpainting or toy feasibility example should be read only as feasibility evidence, not as a deployment claim. Diffusion, flow, and score-based priors should be evaluated under the same range-null audit and the same negative tests [CITE]. Finally, the posterior-calibration results indicate that sampling quality is bounded by the base estimator’s null-space accuracy, not only by the diversity mechanism.

In conclusion, range-null separability gives a compact language for measured and unmeasured content in ghost imaging. It enables a plug-in, test-time, ground-truth-free certificate of measurement consistency, explains when that certificate can or cannot improve image quality, and exposes the fundamental boundary of verification: consistency is certifiable, but the correctness of prior-supplied null-space content is not certified by the measurement alone.

APPENDIX A: FULL POSTERIOR-SAMPLING DIAGNOSTICS

Table XIV gives the full gate-by-gate posterior-sampling diagnostics summarized in Table VI.

TABLE XIV

FULL POSTERIOR-SAMPLING DIAGNOSTICS. THE ROW-SPACE-ONLY DIAGNOSTIC RESTORES P_0 VARIATION WHILE PRESERVING MEASUREMENT CONSISTENCY, BUT COVERAGE REMAINS UNDER-CALIBRATED. ENTRIES MARKED N.R. WERE NOT RECORDED FOR THAT GATE OR ARE NOT APPLICABLE TO THE CALIBRATION/BOTTLENECK DIAGNOSTIC REPRESENTED BY THE ROW.

Gate	diagnostic	fixed- y std	P_0 var	max RelMeasErr	reported status
1	deterministic Rad-5 baseline	0.0010	1.09×10^{-6}	3.84×10^{-5}	collapse
2	row-space-only reconstruction	0.0335	1.28×10^{-3}	5.61×10^{-5}	anti-collapse pass
3	spectral sanity	0.0335	1.28×10^{-3}	5.61×10^{-5}	slope -2.62; high/low 5.37×10^{-3}
4	uncentered calibration audit	n.r.	n.r.	n.r.	90% pixel/ P_0 cov. 0.381 / 0.407; offset 0.0549
5	centered diversity	0.0069	5.19×10^{-5}	4.63×10^{-5}	90% pixel/ P_0 cov. 0.106 / 0.102; offset 0.0180
6	anchor/diversity scan	0.0349	1.39×10^{-3}	5.62×10^{-5}	90% pixel/ P_0 cov. 0.453 / 0.478; offset 0.0331
7	base P_0 bottleneck	n.r.	n.r.	n.r.	det-to-GT RMSE 0.0810; best mean-to-GT RMSE 0.0877